Lab 2: Claude re-asks because the tool_result never came back

Topic 1: Agentic Architecture & Orchestration · Task 1.1: Design and implement agentic loops for autonomous task execution

Runs on Simulator 2: Code Generation with Claude Code · Self-practice

Source: SkillCertPro practice exam Part 1, Q17 (unofficial third-party question)

The question

A workflow script receives a response with stop_reason: "tool_use" for a client-executed run_test tool. The script runs the test successfully, but Claude keeps asking to run the same tool again instead of interpreting the results. What should the script change in its next request?

A. Append a user text summary first, then include the matching tool_result block later in that same turn.

B. Append a user text turn with the test output, then resume the conversation with a normal user instruction.

C. Append the assistant tool_use turn, then add an immediate user turn with a matching tool_result block.

D. Append an assistant text turn with the test output, then add a user turn asking Claude to continue.

Simulate it live

```
# In the browser: open Simulator 2, then Lab 2
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 2

# In a terminal, from simulators/simulator-2-code-generation/
uv run casekit.py --case UC-17
```

What you should see

You draw the three turns: prompt, tool_use with an id, tool_result with that same id.

Append the assistant turn that contains the tool_use block, then a user turn whose first content block is a tool_result with the matching tool_use_id. Text summaries or a wrong role break the link and Claude re-requests the tool.

Open `simulators/simulator-2-code-generation/evidence/` and find it yourself.

The answer

C. Append the tool_use turn, then a user turn with a matching tool_result.

Claude does not run your local tool. It returns a structured tool_use request, your application runs it, and the next request contains the original conversation, the assistant message with the tool_use block, and an immediately following user message with the matching tool_result . The conversation history is the state. The result's tool_use_id ties it to the request's id . tool_result blocks belong in a user message that immediately follows the assistant's tool_use message, and they must come before any text in that message.

Why the others fail

A. Text placed before the tool_result breaks the required ordering: results must come first.

B. Plain text is not linked to the tool_use id, so Claude still sees an unanswered request.

D. Wrong role. The output is presented as Claude's own words, and there is no tool_result at all.

Check yourself

- learner states that tool_result goes in a user message immediately after the assistant tool_use message
- learner names tool_use_id as the link between request and result

Now change one thing

Set LAB_MODEL=claude-sonnet-5-5, rerun, and compare the result and the cost line in evidence/.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.